XPS characterization of MoS2@SiO2 core-shell structures

"What XPS peaks confirm SiO2 coating on MoS2" *Generated by LeapSpace*

SiO_2 coating on MoS_2 is most directly confirmed by the appearance of a Si 2p peak near 103 eV together with an O 1s peak near 532.8 eV, while the Mo 3d doublet for MoS_2 remains present. In the retrieved evidence, **Si 2p around 103 eV** is assigned to **SiO_2**, and **O 1s at 532.8 eV** is assigned to the **Si–O–Si unit**; the MoS_2 component is still identified by **Mo 3d5/2 at 228.5 eV** and **Mo 3d3/2 at 231.6 eV** [69] [70].

Peaks to look for

XPS region	Peak position(s) reported	What it indicates
Si 2p	~103 eV	SiO_2 formation / silica coating [69]
O 1s	~532.8 eV	Si–O–Si in silica [70]
Mo 3d	228.5 eV and 231.6 eV	MoS_2 core remains present [69]

Practical interpretation

- If you see **new Si 2p intensity at ~103 eV**, that is the clearest XPS sign of a **SiO_2 shell/coating** on MoS_2 [69].
- If that Si 2p signal is accompanied by **O 1s at ~532.8 eV**, it strengthens the assignment to **silica rather than generic oxygen contamination** [70].
- The retained **Mo 3d MoS_2 doublet** supports that the substrate/core is still MoS_2 after coating [69].

Bottom line: the key confirming XPS features are **Si 2p at ~103 eV** and **O 1s at ~532.8 eV**, with **Mo 3d peaks at 228.5/231.6 eV** showing the MoS_2 backbone is still intact [69] [70].

Confidence level: **High** — the retrieved evidence directly assigns these binding energies to SiO_2-related species and MoS_2 core signals.

References

69. Xu, Y., Xie, M., Li, Y., (...), Zhu, M. (2020) The effect of Si content on the structure and tribological performance of MoS2/Si coatings. *Surface and Coatings Technology*. https://doi.org/10.1016/j.surfcoat.2020.126362

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